

JR Analyst Python Home Exam

You're a Junior Data Analyst given New York City crime data. You're asked to explore patterns, find trends, and predict the next crimes in town. You'll use the python libraries `pandas` and `matplotlib` for data manipulation and visualizations.

Please complete this exercise and document your process using a **Jupyter Notebook**, with clearly separated code and markdown sections. When you finish the task, send the .ipynb file as your answer, along with your dataset if you made changes to it.

Dataset - `nyc_crime_large.csv`

Your CSV file contains the following columns:

Column	Description
<code>CMPLNT_FR_DT</code>	Date the crime occurred
<code>OFNS_DESC</code>	Offense description
<code>BORO_NM</code>	Borough (e.g., MANHATTAN, BROOKLYN)
<code>LAW_CAT_CD</code>	Crime severity: <code>FELONY</code> , <code>MISDEMEANOR</code> , or <code>VIOLATION</code>
<code>Latitude</code>	Latitude coordinate of the crime
<code>Longitude</code>	Longitude coordinate of the crime

The data comes from an old system with very little governance and monitoring, so expect the data to be a bit messy.

Tasks (To Be Completed in the Notebook)

- Load the data into a pandas dataframe
 - Create two new columns: `year` and `month` from the `CMPLNT_FR_DT` column.
 - Look through the data, find where it is messy and clean it.
- Following are the first questions you need to answer as a Junior Analyst. Remember, if you find problems in the data, clean them first before you answer.
 - What are the top 5 most common crime types?
 - How has the total number of crimes changed over the years?
 - Which 3 boroughs have the most crimes?
 - What's the distribution of crime severity?
- Visualizations - use `matplotlib` to add graphs for each question in Q2